



```
In [55]: import pandas as pd
# Mysql Toolkit
import pymysql # This will work as adapter
from sqlalchemy import create_engine
import psycopg2
```

```
In [56]: df = pd.read_csv(r"C:\Users\shash\OneDrive\Desktop\project\Walmart.csv")
```

```
In [57]: df.head()
```

```
Out[57]:
```

	invoice_id	Branch	City	category	unit_price	quantity	date	time
0	1	WALM003	San Antonio	Health and beauty	\$74.69	7.0	05/01/19	13:08:00
1	2	WALM048	Harlingen	Electronic accessories	\$15.28	5.0	08/03/19	10:29:00
2	3	WALM067	Haltom City	Home and lifestyle	\$46.33	7.0	03/03/19	13:23:00
3	4	WALM064	Bedford	Health and beauty	\$58.22	8.0	27/01/19	20:33:00
4	5	WALM013	Irving	Sports and travel	\$86.31	7.0	08/02/19	10:37:00

```
In [58]: df.tail()
```

```
Out[58]:
```

	invoice_id	Branch	City	category	unit_price	quantity	date	time
10046	9996	WALM056	Rowlett	Fashion accessories	\$37	3.0	03/08/23	13:08:00
10047	9997	WALM030	Richardson	Home and lifestyle	\$58	2.0	22/02/21	10:29:00
10048	9998	WALM050	Victoria	Fashion accessories	\$52	3.0	15/06/23	13:23:00
10049	9999	WALM032	Tyler	Home and lifestyle	\$79	2.0	25/02/21	10:37:00
10050	10000	WALM069	Rockwall	Fashion accessories	\$62	3.0	26/09/20	10:37:00

```
In [59]: df.shape
```

```
Out[59]: (10051, 11)
```

```
In [60]: df.describe()
```

```
Out[60]:
```

	invoice_id	quantity	rating	profit_margin
count	10051.000000	10020.000000	10051.000000	10051.000000
mean	5025.741220	2.353493	5.825659	0.393791
std	2901.174372	1.602658	1.763991	0.090669
min	1.000000	1.000000	3.000000	0.180000
25%	2513.500000	1.000000	4.000000	0.330000
50%	5026.000000	2.000000	6.000000	0.330000
75%	7538.500000	3.000000	7.000000	0.480000
max	10000.000000	10.000000	10.000000	0.570000

```
In [61]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10051 entries, 0 to 10050
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   invoice_id            10051 non-null  int64
1   Branch                10051 non-null  object
2   City                  10051 non-null  object
3   category              10051 non-null  object
4   unit_price            10020 non-null  object
5   quantity              10020 non-null  float64
6   date                  10051 non-null  object
7   time                  10051 non-null  object
8   payment_method        10051 non-null  object
9   rating                10051 non-null  float64
10  profit_margin         10051 non-null  float64
dtypes: float64(3), int64(1), object(7)
memory usage: 863.9+ KB
```

```
In [62]: df.duplicated().sum()
```

```
Out[62]: np.int64(51)
```

```
In [63]: df.drop_duplicates(inplace=True)
```

```
In [64]: df.duplicated().sum()
```

```
Out[64]: np.int64(0)
```

```
In [65]: df.shape
```

```
Out[65]: (10000, 11)
```

```
In [66]: df.isnull().sum()
```

```
Out[66]: invoice_id      0  
Branch      0  
City        0  
category    0  
unit_price  31  
quantity    31  
date        0  
time        0  
payment_method  0  
rating      0  
profit_margin  0  
dtype: int64
```

```
In [67]: df.dropna(inplace=True)
```

```
In [68]: df.isnull().sum()
```

```
Out[68]: invoice_id      0  
Branch      0  
City        0  
category    0  
unit_price  0  
quantity    0  
date        0  
time        0  
payment_method  0  
rating      0  
profit_margin  0  
dtype: int64
```

```
In [69]: df.dtypes
```

```
Out[69]: invoice_id      int64  
Branch      object  
City        object  
category    object  
unit_price  object  
quantity    float64  
date        object  
time        object  
payment_method  object  
rating      float64  
profit_margin  float64  
dtype: object
```

```
In [70]: df["unit_price"] = df["unit_price"].str.replace("$", "", regex=False).astype(f
```

```
In [71]: df.dtypes
```

```
Out[71]: invoice_id      int64  
Branch      object  
City        object  
category     object  
unit_price  float64  
quantity    float64  
date        object  
time        object  
payment_method object  
rating      float64  
profit_margin float64  
dtype: object
```

```
In [72]: df.head()
```

```
Out[72]:
```

	invoice_id	Branch	City	category	unit_price	quantity	date	time
0	1	WALM003	San Antonio	Health and beauty	74.69	7.0	05/01/19	13:08:00
1	2	WALM048	Harlingen	Electronic accessories	15.28	5.0	08/03/19	10:29:00
2	3	WALM067	Haltom City	Home and lifestyle	46.33	7.0	03/03/19	13:23:00
3	4	WALM064	Bedford	Health and beauty	58.22	8.0	27/01/19	20:33:00
4	5	WALM013	Irving	Sports and travel	86.31	7.0	08/02/19	10:37:00

```
In [73]: df["date"] = df["date"].astype("datetime64[ns]")
```

```
In [74]: df["Month"] = pd.to_datetime(df["date"]).dt.month
```

```
In [75]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9969 entries, 0 to 9999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   invoice_id            9969 non-null   int64
1   Branch                9969 non-null   object
2   City                  9969 non-null   object
3   category              9969 non-null   object
4   unit_price            9969 non-null   float64
5   quantity              9969 non-null   float64
6   date                  9969 non-null   datetime64[ns]
7   time                  9969 non-null   object
8   payment_method        9969 non-null   object
9   rating                9969 non-null   float64
10  profit_margin         9969 non-null   float64
11  Month                  9969 non-null   int32
dtypes: datetime64[ns](1), float64(4), int32(1), int64(1), object(5)
memory usage: 973.5+ KB
```

```
In [78]: df['Total_Price'] = df['unit_price'] * df['quantity']
```

```
In [79]: # Mysql Connection
user = "root"
password = "shashank123"
db = "walmart_db"
```

```
In [80]: #Mysql connection
engine_mysql = create_engine("mysql+pymysql://root:shashank123@localhost:3306/
try:
    engine_mysql
    print("Mysql Connection Successful")
except:
    print("Mysql Connection Failed")
```

Mysql Connection Successful

```
In [81]: df.columns = df.columns.str.lower()
```

```
In [82]: df.columns
```

```
Out[82]: Index(['invoice_id', 'branch', 'city', 'category', 'unit_price', 'quantity',
               'date', 'time', 'payment_method', 'rating', 'profit_margin', 'month',
               'total_price'],
              dtype='object')
```

```
In [84]: select * from walmart;
```

```
Cell In[84], line 1
    select * from walmart;
           ^
SyntaxError: invalid syntax
```

In []: